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Production Improvement of Multi-Fractured Horizontal Wells in Unconventional Resources Using Fracture Propagation and Numerical Simulation Techniques

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Abstract

To investigate the impact of actual fracture propagation morphology on productivity prediction in multi-cluster fracturing and optimize stimulation designs, this study employs the displacement discontinuity method (DDM) to analyze the effects of cluster spacing and cluster number on fracture propagation patterns and fluid distribution. Using CMG numerical simulation software with actual geological parameters from Block H of Jilin Oilfield, a homogeneous single-well fracturing model was established to determine optimal stimulation parameters for maximum productivity. Results indicate that as cluster numbers increase, the range of fluid intake percentage and fracture length differences initially expands. Challenges such as unclear fracture propagation patterns and frequent well interference significantly constrain efficient development of low-permeability reservoirs. Through integration of geological characteristics and fracture propagation simulations, the study analyzes volumetric fracture propagation in horizontal well groups and the influence of natural fractures on hydraulic fracture growth, clarifying key well interference factors. Key findings include: (1) In naturally fractured zones, hydraulic fracture behavior is dominated by the induction, capture, and barrier effects of natural fractures; (2) Asymmetric propagation and non-uniform initiation caused by natural fractures are primary drivers of well interference; (3) The development degree of natural fractures critically influences hydraulic fracture propagation. These insights provide a scientific basis for optimizing hydraulic fracturing parameters in the Changning shale gas reservoir.

Introduction

Hydraulic fracturing is a key technology for efficiently developing unconventional resources such as shale gas, shale oil, and tight oil/gas reserves. Understanding the distribution patterns of proppants within hydraulic fractures is fundamental for optimizing sand-laden fracturing, creating high-conductivity flow channels. To investigate proppant distribution and settling within fractures, extensive research has been conducted globally through laboratory physical models and numerical simulations. Most models were based on the simplified KGD fracture model. The fracture was simplified as a rectangular prism, omitting variations in fracture width and height along the length. Early studies used parallel glass plates to model fractures and studied proppant distribution and settling behavior, observing the formation of sand beds as proppants settled after entering fractures, with subsequent proppants moving deeper into the fracture driven by the fluid. Researchers such as Malhotra and Wen Qingzhi utilized large, visual plate setups to simulate artificial fractures to exam how factors like fluid viscosity, fracture width, and sand-to-fluid ratios impact proppant distribution and settling, and established sensitivity analyses for these variables. Besides these factors, they also studied the impact of perforation parameters and fracture morphology. During volume fracturing, fractures often form as networks, and Wen Qingzhi and others used complex "T"- and "H"-type models to study proppant movement within these networks, proposing that dune formation in branch fractures is the result of both sand carrying force and gravity. Low-permeability hydrocarbon reserves represent substantial global resources, driving worldwide focus on horizontal well multi-cluster fracturing as a key recovery technology. Field observations reveal that stress interference during simultaneous fracture propagation suppresses extension at perforation clusters, limiting stimulation effectiveness in tight reservoirs. Dynamically altered stress fields critically govern fracture trajectories and fluid distribution. Prior research established foundational insights: Fisher et al. (Barnett Shale) correlated increased production with larger fluid volumes and fracture lengths; Buffington (Monterey Shale) demonstrated significant recovery improvements via staged multi-cluster fracturing in nanodarcy-permeability zones; Nijhoff emphasized the necessity of adequate fracture network conductivity alongside large stimulated reservoir volumes (SRV). Olson's pseudo-3D boundary element model (2009) simulated non-planar fracture growth using displacement discontinuity method (DDM) but neglected fluid-rock coupling, later partially resolved by Wu Kan through fluid flow integration. Gao, Y., Rahman, M.M., and Lu, J (2021) transient pressure analysis of multi-fractured horizontal well in naturally-fractured oil reservoir , Al Hameli, F., Suboyin, A., Al Kobaisi, M., Rahman, M.M., Haroun, M. (2022) build pseudo-3D-carter-dual-porosity model in a dual-porosity system. Domestically, Xu Chuangchao optimized fracture parameters via Eclipse-based modeling incorporating wellbore friction and threshold pressure gradients, while Liu Xiong analyzed productivity influencers using dual-porosity models. Current optimizations remain constrained by idealized uniform-fracture assumptions, disregarding actual geometries. Addressing this gap, our study employs DDM to quantify cluster spacing/number effects on propagation patterns and fluid distribution, then establishes a CMG-based homogeneous model using Block B (Xinjiang Oilfield) geological parameters. By importing actual fracture geometries under varied operational parameters and conducting productivity-driven optimization, we determine field-applicable parameters for maximum production.

Multi-Fracture Propagation Laws

Fractures, ubiquitous structural features in rock formations, critically impact subsurface stability and engineering outcomes. In petroleum extraction, fracture initiation and propagation compromise reservoir integrity, adversely affecting hydrocarbon productivity. Traditional characterization oversimplifies fractures as geometrically regular surfaces, neglecting intrinsic complexity. Fractal theory reveals their inherent self-similarity and scale invariance, enabling accurate morphological and evolutionary modeling. This paradigm shift facilitates robust rough fracture models that replicate true fracture networks through fractal algorithms, capturing structural heterogeneity and distribution patterns. Such models provide critical insights into fracture mechanics and growth prediction while supporting engineering applications: they quantify proppant transport efficiency in hydraulically fractured wells by simulating fracture connectivity and spatial configuration, optimize stimulation parameters, and derive fracture porosity/permeability to predict reservoir flow behavior and production capacity. Ultimately, fractal-based modeling advances fracture diagnostics and stimulation design for enhanced recovery.

During multi-cluster fracturing in horizontal wells, cluster-wise fluid distribution is a dynamic process where injection rates into each fracture evolve over time. Given the critical influence of fluid influx on hydraulic fracture dimensions, a dynamic fluid allocation model must be established. As shown in Figure 1, wellbore flow adheres to Kirchhoff's law, requiring the total injection rate to satisfy:

$$Q_T = \sum_{i=1}^{m} Q_i \tag{1}$$

Where Q_T = total injection rate, Q_i = injection rate into fracture i, and m = number of fractures.

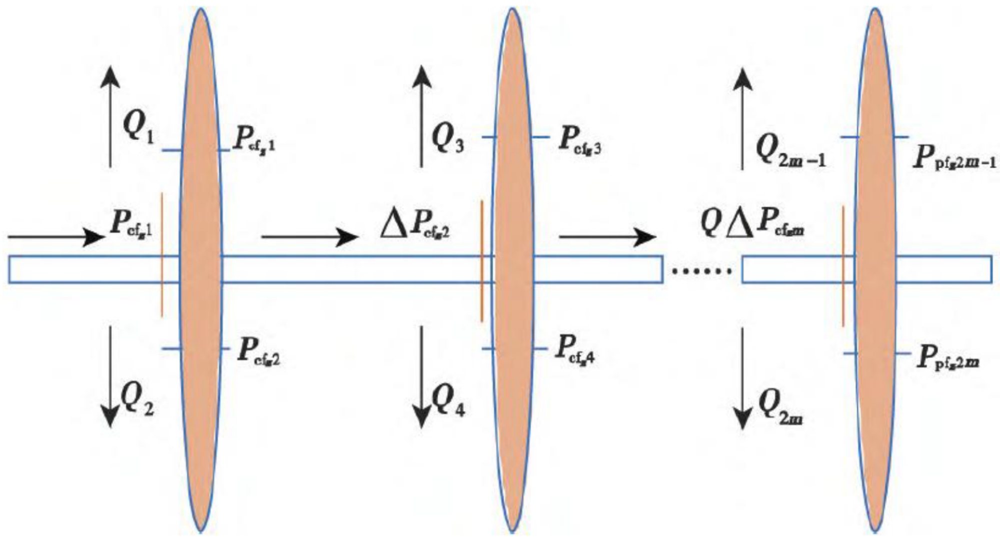

Figure 1—Schematic of Multi-Cluster Fracturing Fluid Flow in Horizontal Well

Accounting for wellbore and perforation pressure drops, wellbore flowing pressure follows:

$$p_0 = p_{w,i} + p_{pf,i} + p_{cf,i} \tag{2}$$

where p_0 = wellhead flowing pressure, $p_{w,i}$ = fracture entrance pressure, $p_{fp,i}$ = perforation friction pressure drop, and
$p_{cf,i}$ = wellbore friction pressure drop. Perforation friction is calculated as:

$$p_{pf,i} = \frac{2.2326 \times 10^{-10} Q_i^2 \rho}{n_p^2 d^4 C^2} \tag{3}$$

Wellbore friction is given by:

$$p_{cf,i} = 2^{3n+2} \pi^{-n} K \left(\frac{1+3n}{n} \right) D - (3n+1) \sum_{j=1}^{i} Q_{w,j}^n L_{w,j} \tag{4}$$

where n_p = perforation count, d = perforation diameter, C = discharge coefficient, ρ = fluid density, D = wellbore diameter, $Q_{w,j}$ j= flow rate between clusters j−1 and j, and $L_{w,j}$ = cluster spacing.

The system and individual fractures must satisfy mass balance equations:

$$\int_0^T Q_T(t)dt = \sum_{i=1}^{2m} \left[2Hc_t \int_0^{f,t} \int_0^T \frac{dsdt}{\sqrt{t - \tau(s)}} + \int_0^{\mathrm{fi},t} Hwds \right] \tag{5}$$

$$\begin{cases}\int_0^T Q_1(t)dt = 2Hc_t\int_0^{f1,t}\int_0^T \frac{dsdt}{\sqrt{t-\tau(s)}} + \int_0^{f1,t} Hwds \\ \int_0^T Q_2(t)dt = 2Hc_t\int_0^{f2,t}\int_0^T \frac{dsdt}{\sqrt{t-\tau(s)}} + \int_0^{f2,t} Hwds \\ \int_0^T Q_i(t)dt = 2Hc_t\int_0^{f,t}\int_0^T \frac{dsdt}{\sqrt{t-\tau(s)}} + \int_0^{f,i,t} Hwds\end{cases} \tag{6}$$

With wellhead pressure known at each timestep, fracture inflow rates (Qi), internal flow distribution (qi), and timestep size are solved iteratively:

$$Q_{i,j+1} = \left(1-\alpha_1\right)Q_{i,j} + \alpha_1 Q_{i,j+1/2} \tag{7}$$

$$q(i)_{k,j+1} = \left(1-\alpha_2\right)q(i)_{k,j} + \alpha_2 q(i)_{k,j+1/2} \tag{8}$$

Iteration terminates when flow rates converge:

$$\begin{cases}\dfrac{\sum_{i=1}^{2m}\left|Q_{i,j+1}-Q_{i,j}\right|}{\sum_{i=1}^{2m}Q_{i,j+1}} < \lambda \\ \dfrac{\sum_{i=1}^{2m}\sum_{k=1}^{m_{ki}}\left|q(i)_{k,j+1}-q(i)_{k,j}\right|}{\sum_{i=1}^{2m}\sum_{k=1}^{m_{ki}}q(i)_{k,j+1}}\end{cases} \tag{9}$$

where $Q_{i,j+1}$ = flow rate at fracture, iteration j; $q_{(i)k,j+1}$ = flow at node k of fracture i; α_1, α_2 =0.1 (iteration factors); λ = convergence tolerance; and m_{ki} = nodal count per fracture.

Gas Injection Scheme Design

3D Geological Modeling for Fractured Reservoirs

Based on conventional logging data and imaging data, a reservoir geomechanical model is established. The model parameters are calibrated using data from drilling, formation testing, fracturing tests, and core experiments (Figure 2) . The research results are used to guide the mud weight design for the drilling engineering plan of the Y102H36 platform.

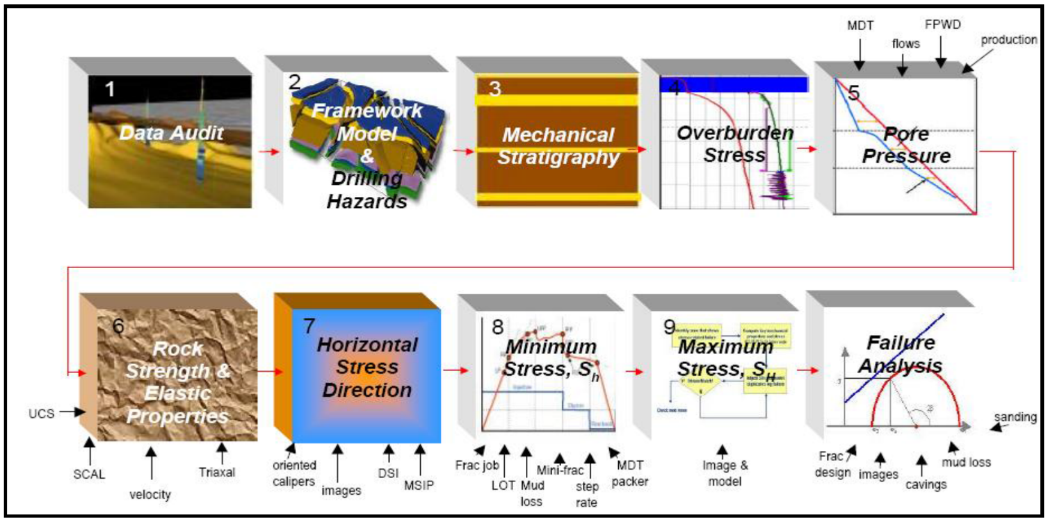

Figure 2—Geomechanical Model Workflow

Rock Mechanics Modeling

Dynamic elastic modulus is calculated using the P-wave and density logs. The elastic modulus is referred to as dynamic because the acoustic logging is conducted at high frequencies (approximately 10 kHz). Compared to mechanical property testing in laboratories or under stable loads at the surface/near the wellbore, rocks exhibit stiffer characteristics under dynamic conditions. The dynamic elastic modulus and Poisson's ratio are calculated using the following correlation:

$$E_{dyn} = \rho V_s^2\left(3V_p^2 - 4V_s^2\right)/\left(V_p^2 - V_s^2\right) \tag{10}$$

$$\nu_{dyn} = \left(V_p^2 - 2V_s^2\right)/2\left(V_p^2 - V_s^2\right) \tag{11}$$

Elastic Modulus. The elastic modulus E of a rock refers to the ratio of axial stress to the increment of axial strain under uniaxial compression conditions, as shown in the following equation:

$$E = \Delta\delta_z / \Delta\varepsilon_z \tag{12}$$

In this equation, E—the elastic modulus—is the slope of the stress-strain curve, representing the rate of change of stress with respect to strain under uniaxial stress conditions.

$\Delta\delta_z$, $\Delta\varepsilon_z$—increments of axial stress and strain.

It has a significant impact on fracture width and fracturing pressure. In a two-dimensional model where the fracture height is assumed to be constant, for Newtonian fluids, the fracture width is inversely proportional to the fourth root of the elastic modulus, i.e., $W \sim 1/E^{1/4}$; for non-Newtonian fluids, the relationship between fracture width and elastic modulus is $W \sim 1/E^{1/2n'+2}$, where n' is the power-law index of the non-Newtonian fluid. Whether in a two-dimensional model or a three-dimensional model, calculations show that as the elastic modulus of the rock increases, the fracture width decreases.

According to the two-dimensional fracture model, for Newtonian fluids, the net pressure during fracturing operations is proportional to the 3/4 power of the elastic modulus, i.e., $\Delta p(t) \sim E^{3/4}$; for non-Newtonian fluids, the relationship between pressure and elastic modulus is $\Delta p(t) \sim E^{(2n'+1)/(2n'+2)}$. As the elastic modulus increases, the net pressure during the operation will also increase.

Poisson's ratio. The Poisson's ratio of a rock refers to the ratio of lateral strain to axial strain under uniaxial compression conditions.

$$\nu = -\frac{\varepsilon_r}{\varepsilon_z} \tag{13}$$

Poisson's ratio is an important parameter in determining horizontal ground stress. Generally, due to the relationship between lateral strain and axial strain, as well as the relationship between axial strain and axial stress, which are often in the form of curves, the Poisson's ratio of rock tends to increase with increasing axial stress. The linear segment of the stress-strain curve is commonly used as the basis for calculations.

The deformation modulus and Poisson's ratio of a rock mass are influenced by various factors, including the mineral composition, structure and texture of the rock, degree of weathering, porosity, moisture content, microstructural surfaces, and their relationship with the load direction, leading to significant variability.

Shear Modulus. The shear modulus is the ratio of shear stress to strain within the elastic deformation limit under shear stress, also known as the modulus of rigidity or rigidity modulus. It is one of the mechanical property indicators of a material, representing the material's ability to resist shear strain. A larger modulus indicates stronger rigidity, making the rock more resistant to shear deformation; when the shear modulus of a rock is infinite, the rock becomes a rigid body. The reciprocal of the shear modulus is called the shear compliance, which measures the amount of shear strain under a unit shear force, indicating the ease of shear deformation of the material.

The shear modulus G of a rock is related to its elastic modulus E and Poisson's ratio v as follows:

$$G = \frac{E}{2(1+v)} \tag{14}$$

Rock Compressive Strength. This refers to the ultimate strength of a rock specimen under uniaxial pressure at the point of failure, numerically equal to the maximum axial stress at failure, usually denoted as σ_e:

$$\sigma_c = \frac{P}{A} \tag{15}$$

In the equation, P represents the applied load at failure, known as the failure load; A is the original cross-sectional area.

If the results from rock mechanics laboratory measurements are available, this correlation will be calibrated by matching the rock mechanical properties derived from well logging with the laboratory measurement results.

Horizontal Stress Modeling

Stress Orientation Analysis

Borehole breakouts represent wellbore intervals experiencing shear failure. During drilling, mud replacement redistributes in-situ stresses, creating localized stress concentrations. In vertical wells, maximum circumferential compressive stress occurs along the Shmin direction. When this stress exceeds formation compressive strength, shear failure causes rock spallation, forming an elliptical borehole with its major axis (breakout orientation) parallel to Shmin. In regions where overburden stress dominates, continuous breakouts develop vertically symmetric features along the minimum horizontal stress direction. Borehole shear failure occurs when circumferential compressive stress exceeds formation strength, causing elliptical boreholes with major axes (breakout orientation) parallel to Shmin and vertically symmetric features in vertical wells. Observed instability in offset Well A (Figure 3) indicates a NW-SE maximum horizontal stress orientation at ≈110°.

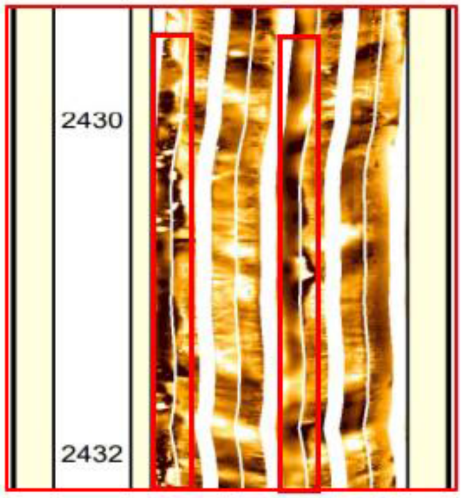

Figure 3—Borehole Breakout Characteristics in Reservoir Section,

Microseismic monitoring of Well A hydraulic fracturing recorded fracture orientations across 13 stages indicating a NW-SE (≈108°) maximum horizontal stress direction consistent with regional trends. Borehole breakout inversion applies Coulomb Fault Theory (Figure 4), where distinct stress regimes correspond to specific Sv-Shmin-SHmax relationships(Brace, 1980): Normal Faulting (NF: Shmin < SHmax < Sv), Strike-Slip (SS: Shmin < Sv < SHmax), and Reverse Faulting (RF: Sv < Shmin < SHmax). Polygonal boundaries represent frictional equilibrium states common in crustal formations. The theory assumes fault slip occurs when shear-to-effective-normal stress ratio exceeds 0.6 friction coefficient (Byerlee, 1978). SHmax magnitude is efficiently constrained using borehole rock strength or observed breakout dimensions, with red contours indicating feasible SHmax ranges derived from Unconfined Compressive Strength (UCS) and breakout width at given depths.

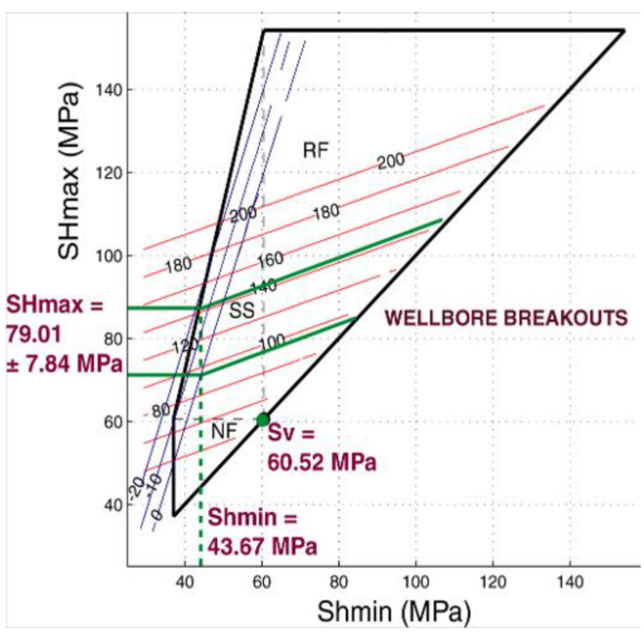

Figure 4—Simulation Methodology for Maximum Horizontal Stress (Shmax)

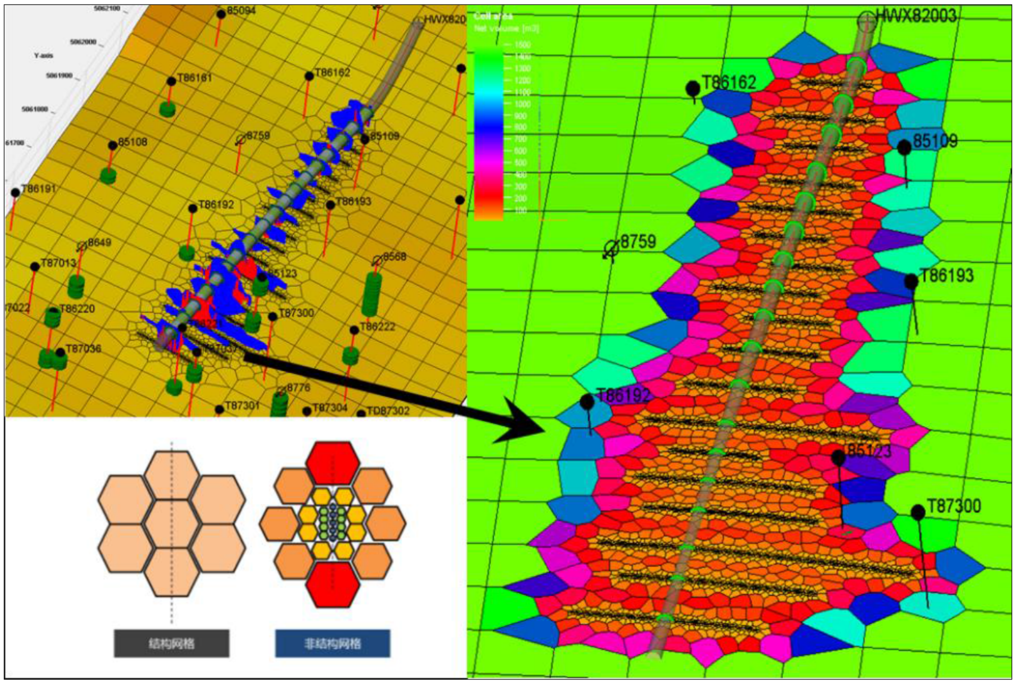

Figure 5—Unconventional Fracture Model (UFM) Mesh for Hydraulic Fractures

Shmin and SHmax magnitudes derived from well logs are estimated using a poroelastic model:

$$S_{h\min} = \frac{v}{1-v} S_V + \frac{1-2v}{1-v} \alpha P_p + \frac{E}{1-v2} \varepsilon_h + \frac{vE}{1-v2} \varepsilon_H$$

$$S_{H\max} = \frac{v}{1-v} S_V + \frac{1-2v}{1-v} \alpha P_p + \frac{E}{1-v2} \varepsilon_H + \frac{vE}{1-v2} \varepsilon_h$$

Where: v = static Poisson's ratio; α = Biot's coefficient; Pp = pore pressure; E = static Young's modulus; Sv = vertical stress; εh, εH = tectonic strains in Shmin and SHmax directions. The final stress model integrates effective stress ratios with log-based calculations. While Biot's coefficient is uniformly 0.9 regionally, tectonic stress coefficients vary across wells due to structural positioning and reservoir heterogeneity. Modeling indicates a dominant strike-slip stress regime (SHmax > Sv > Shmin), favoring vertical fracture development during reservoir stimulation.An Unconventional Fracture Model (UFM) mesh was constructed to characterize fracture networks by integrating natural fractures and hydraulic fractures with geomechanical principles. This approach captures reservoir heterogeneity and precisely delineates planar/vertical gas migration behavior (see Figure 3).

Engineering plan Optimization

Reservoir engineering methods and numerical simulations determined gas-liquid interface migration velocity at 0.03–0.08 m/d in the Lower Urho Formation (monoclinal structure), accounting for gas cap expansion coefficients (~1.1). Benchmarking global practices reveals that Marathon Oil's flank CO_2 injection in Layer Aa boosted Well A16 production by 900% (400→4,000 BOPD) over three years, concurrently reducing water cut from 90% to 15% and elevating GOR from 2,000 to 7,000 scf/STB, while China's Yaha Field flank injection demonstrated gravity-dominated migration along upper high-permeability zones—contrary to intended bottom-layer targeting—resulting in enriched gas accumulation at the reservoir base due to gravity segregation overpowering diffusion.For the HWX82003 pilot in the Lower Urhoe Formation, an optimized two-phase injection strategy was designed: early-phase implementation combines large-PV gas injection with pressure soaking, followed by production via natural flow/gas-lift for high-GOR/high-rate extraction across 4–5 huff-n-puff cycles; late-phase transitions to flank injection flooding deploy vertical injectors in structural highs and horizontal producers to leverage gas override for enhanced sweep efficiency. Field validation through 2020 nitrogen huff-n-puff trials in Wells T86697/T86701 confirmed dual mechanisms—structural-high drainage (29.2 MPa initial pressure) and lateral displacement—enhancing near-wellbore fluid mobility and effectively mobilizing Type III reservoir oil (Figure 6).

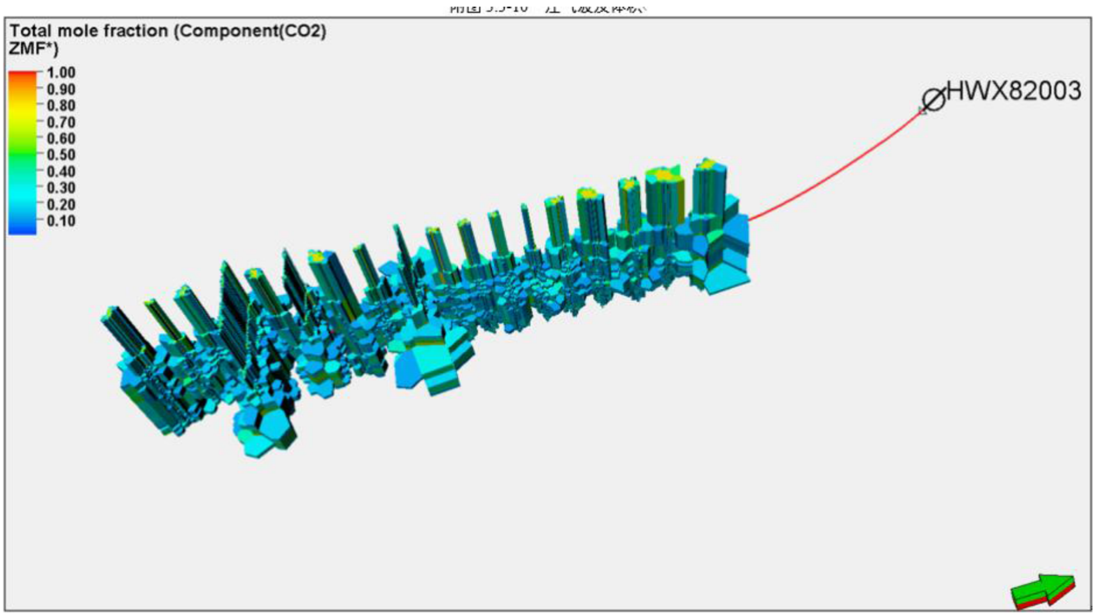

Figure 6—Gas Injection Sweep Volume of Pilot Well Group

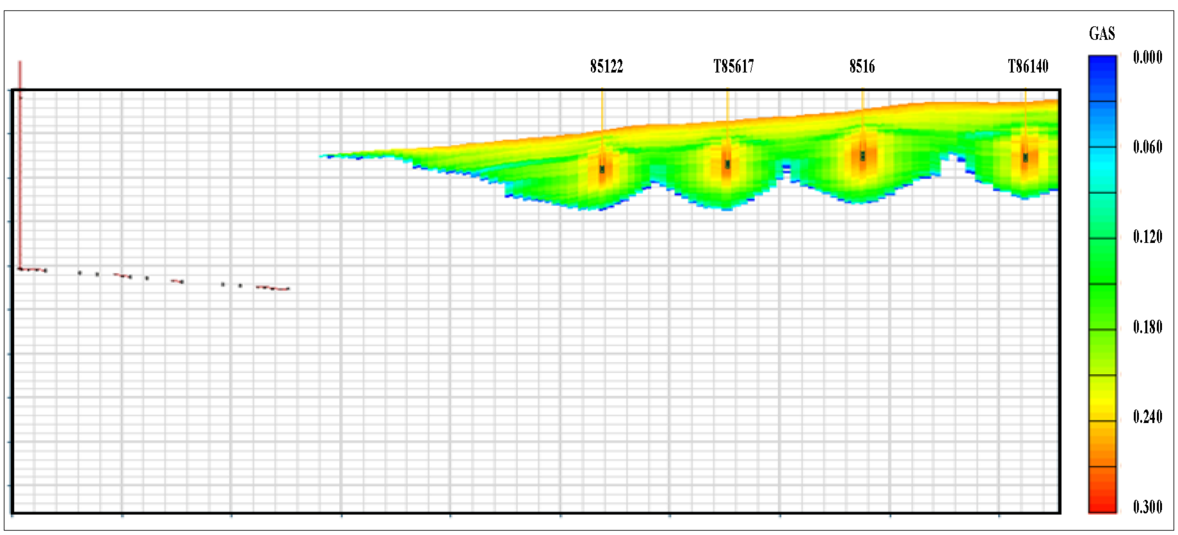

Figure 7—Gas Migration Behavior in the Structural High Injection Zone

Optimized Pilot Scheme Design

Three schemes were comparatively evaluated based on reservoir conditions and operational feasibility:

Table 1—CO_2 Flooding Comparative Scheme Design for HWX82003 Pilot Area

	Baseline	Scheme 1	Scheme 2	Scheme 3
	Depletion Development	CO_2 Huff-n-Puff	Structural High Injection	Flank Injection
Key Features	Aggressive drainage-production	Initial natural flow later pumping-induced flowback Adjacent wells shut in	4 structural-high injectors Production: HW + vertical producers	4 flank injectors Production: HW + vertical producers
Well Pattern	0 injectors 17 producers	Simultaneous injection-production	4 injectors 12 vertical producers 1 horizontal producer	4 injectors 12 vertical producers 1 horizontal producer

(1) Baseline: Depletion current development

Depletion development (0 injectors, 17 producers) yielding 63,000 t cumulative oil and 338,000 t water over 15 years, with 21.5% recovery (2.8% increase from current).

(2) Scheme 1 (Huff-n-Puff):

Scheme 1 (Huff-n-Puff): CO_2 injection (200 t/d per well, 5 cycles × 3,000 t) through HWX82003 horizontal well while shutting adjacent producers. Achieved rapid response but short efficacy: 12,600 t oil and 34,000 t liquid (6.3% recovery).

(3) Scheme 2 (Structural High Injection):

Four vertical injectors (T86140, 85094, 8516, T85617; 50 t/d each) with existing producers. Demonstrated multi-directional sweep but slow response due to gas migration: 178,000 t oil, 606,000 t water, 0.19 t/t displacement efficiency, 27.1% recovery (+5.6% vs baseline).

(4) Scheme 3 (Flank Injection):

Four injectors (8760A, T86194, 8568, T87300; 50 t/d each) perforated in P_2w_1 zone, selective well shut-ins, and pressure monitoring (T86163, T85123). Leveraged gas override for efficient horizontal well drainage: 320,000 t oil, 433,000 t water, 0.34 t/t displacement efficiency, 34.1% recovery (+12.7% vs baseline).Flank gas injection leverages gas override effects to drive incremental migration toward horizontal producers, achieving dual objectives of energy replenishment and oil displacement while reducing water production.Simulation results over a 15-year horizon confirm Scheme 3's superiority: daily injection of 200 t (cumulative 950,000 t) yields 320,000 t cumulative oil and 433,000 t water, with a displacement efficiency of 0.34 t/t and 34.1% recovery—representing a 12.7% increase over baseline performance.Comprehensive evaluation of gas volume, production stability duration, recovery enhancement, and sweep efficiency establishes the optimal strategy: phased implementation of Scheme 1 (huff-n-puff) + Scheme 3 (flank injection).Economic validation for Scheme 1 demonstrates viability under tiered oil pricing: 30-month CO_2 huff-n-puff operation achieves an average input-output ratio of 1.37 and net profit of ¥7.28 million.

Conclusions

In naturally fractured zones, hydraulic fracture propagation is dominated by the induction, capture, and barrier effects of natural fractures: Fracture sets parallel to wellbores restrict growth, causing asymmetric unilateral extension; Diagonally oriented fractures capture and redirect propagation along their planes; Perpendicular fractures induce preferential fracture growthAsymmetric propagation and non-uniform initiation due to natural fractures are primary drivers of well interference.Fracture designs must prioritize the occurrence patterns and developmental intensity of natural fracture networks.A phased injection strategy is recommended: Initial large-PV gas injection followed by pressure soaking, with production via natural flow/gas lift for high-GOR/high-rate extraction over 4–5 huff-n-puff cycles; subsequent transition to flank injection flooding deploys vertical injectors in structural highs and horizontal producers to leverage gas override effects.CO_2 huff-n-puff implementation (200 t/d × 15 days/cycle; 5 cycles) yields 12,600 t cumulative oil and 34,000 t liquid, achieving 6.3% recovery.Flank CO_2 injection over 15 years (cumulative

950,000 t) delivers superior performance: 320,000 t oil, 433,000 t water, 0.34 t/t displacement efficiency, and 34.1% recovery—a 12.7% increase compared with the baseline scenario.drivers.

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